

FPGA1

File: fpga1.kicad_sch

FPGA2

File: fpga2.kicad_sch

Interfaces

File: interfaces.kicad_sch

Audio

File: audio.kicad_sch

Memory

File: memory.kicad_sch

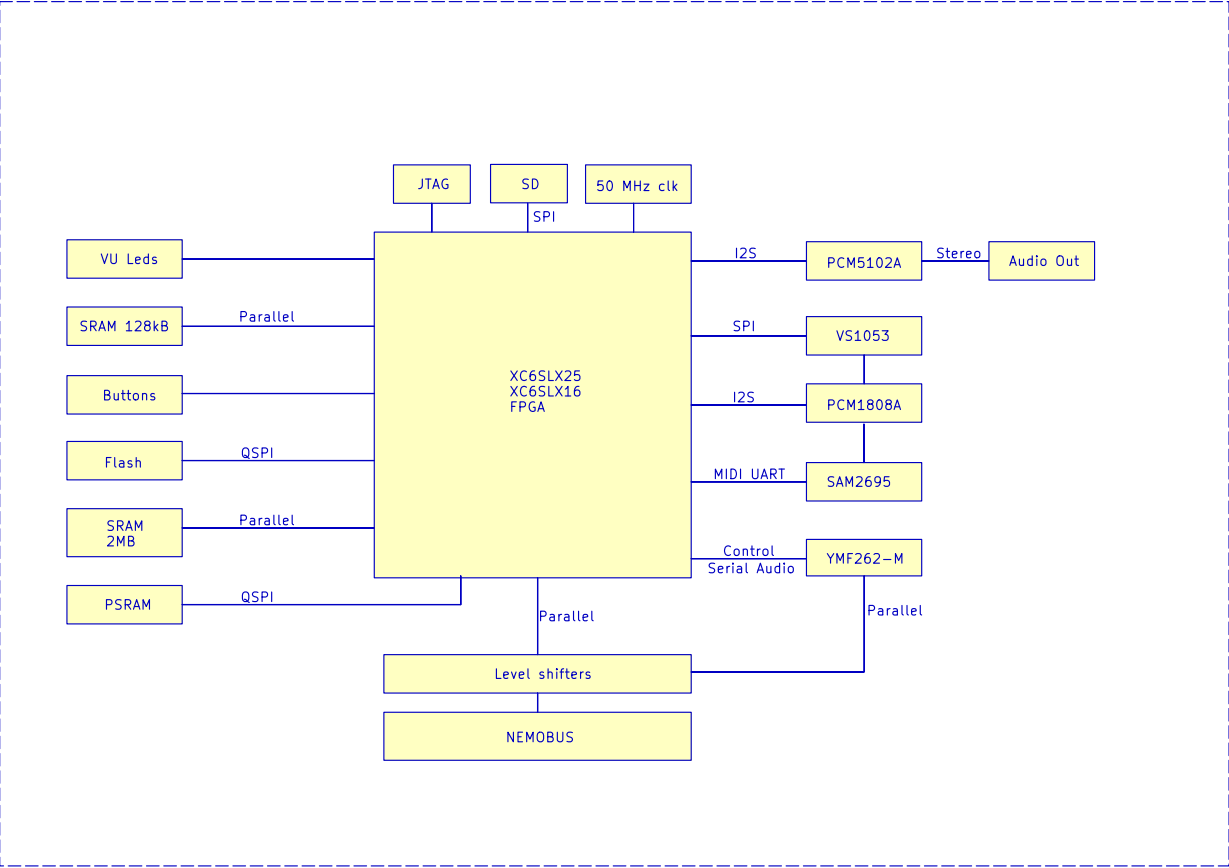
Power

File: power.kicad_sch

VU

File: vu.kicad_sch

FUNCTIONAL DIAGRAM



Sound card for ZX Spectrum (NemoBus)

Andy Karpov

Sheet: /

File: karabas-megabuzz.kicad_sch

Title: Karabas-MegaBuzz

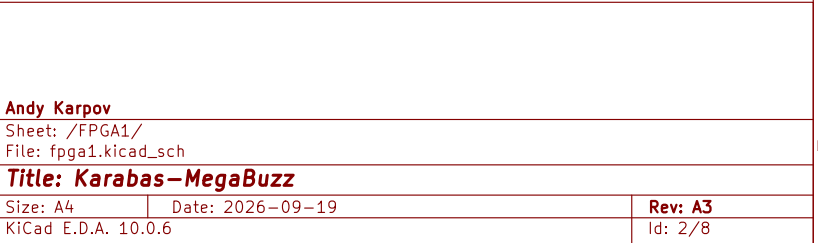
Size: A4

Date: 2026-09-19

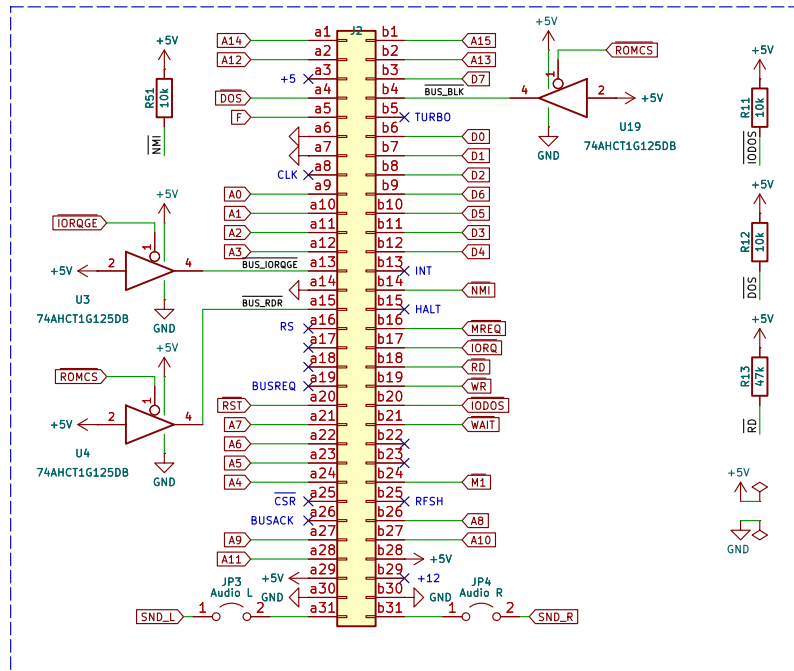
Rev: A3

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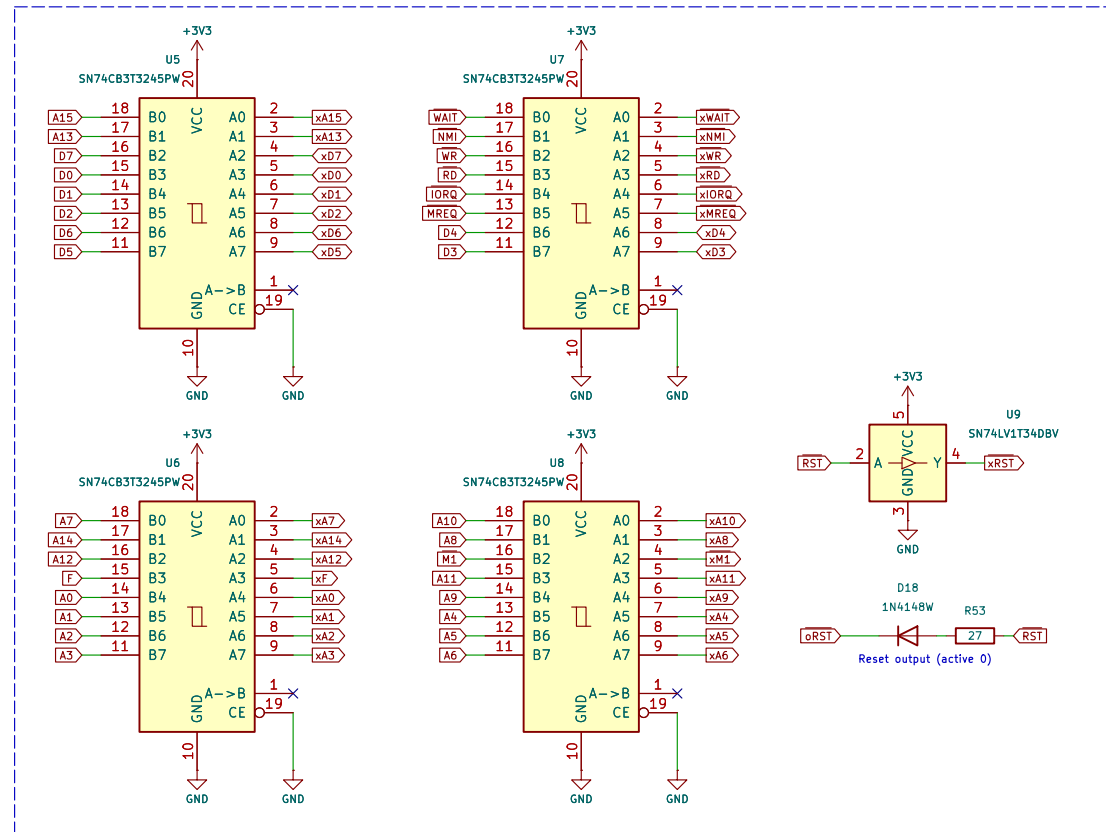
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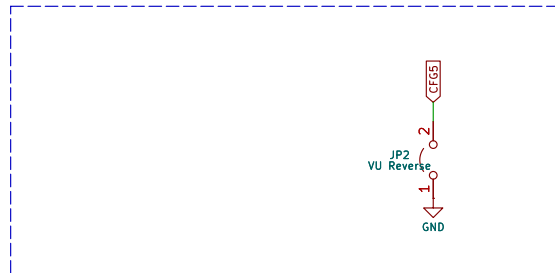
NEMOBUS SLOT



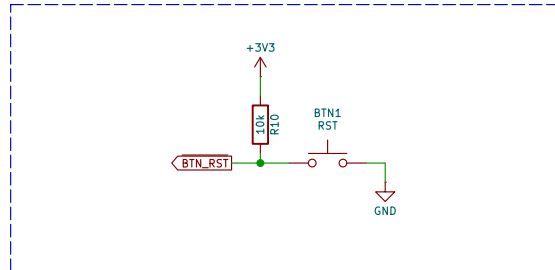
LEVEL SHIFTERS



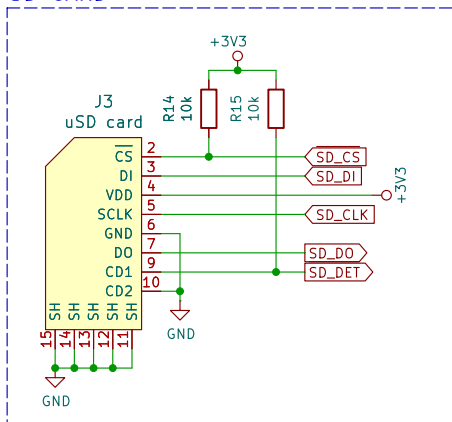
CONFIGURATION



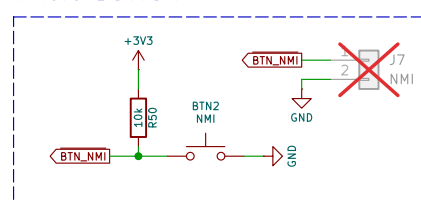
RESET BUTTON



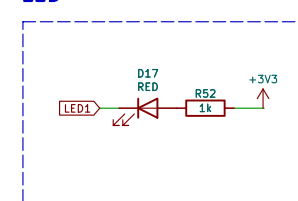
SD CARD



MAGIC BUTTON



LED



Andy Karpov

Sheet: /Interfaces/

File: interfaces.kicad_sch

Title: Karabas-MegaBuzz

Size: A4

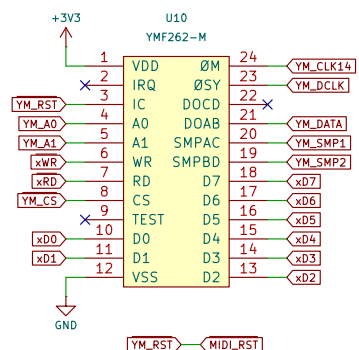
Date: 2026-09-19

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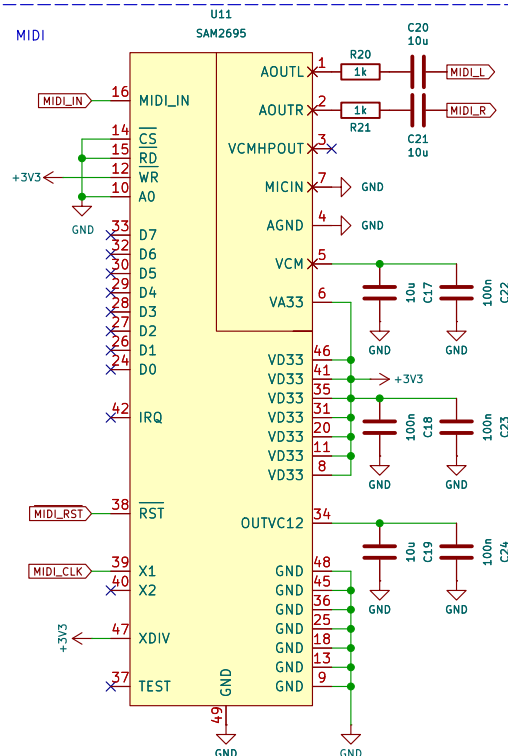
Rev: A3

Id: 4/8

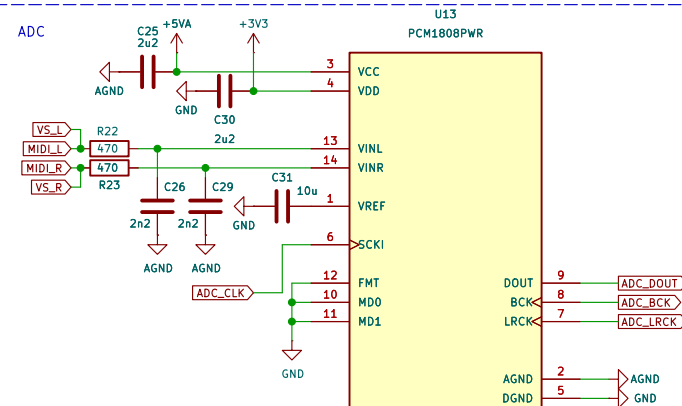
OPL3



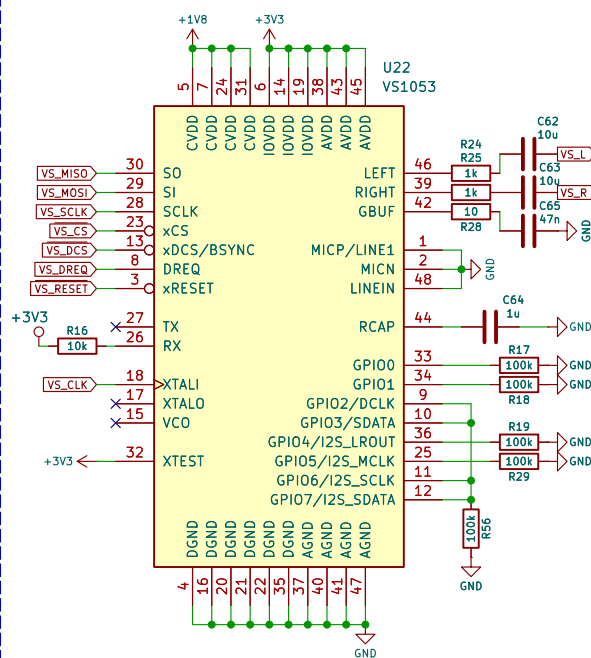
MIDI



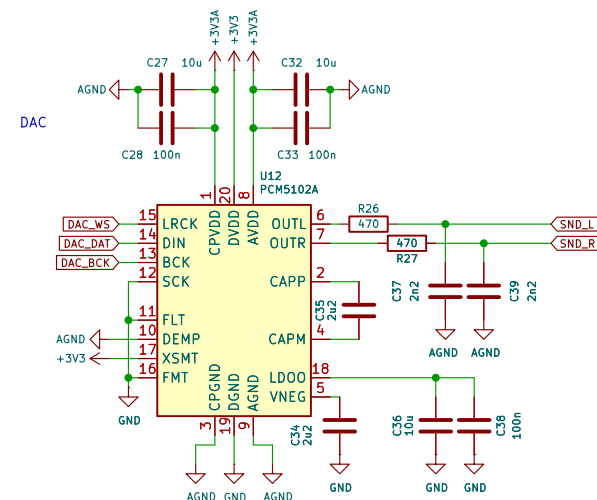
ADC



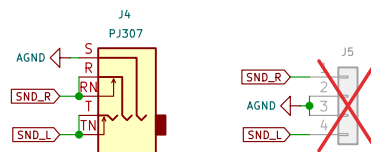
MP3 Decoder



DAC



SOUND OUT



Andy Karpov

Sheet: /Audio/
File: audio.kicad_sch

Title: Karabas–MegaBuzz

Size: A4	Date: 2026-09-19
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Rev: A3

Id: 5/8

U14
CY62167EV30LL

Pin	Label	Pin	Label
25	MA0	29	D0 MD0
24	MA1	31	D1 MD1
23	MA2	33	D2 MD2
22	MA3	35	D3 MD3
21	MA4	38	D4 MD4
20	MA5	40	D5 MD5
19	MA6	42	D6 MD6
18	MA7	44	D7 MD7
8	MA8		
7	MA9		
6	MA10		
5	MA11		
4	MA12		
3	MA13		
2	MA14		
1	MA15		
48	MA16		
17	MA17		
16	MA18		
9	MA19		

Pin 20 is labeled A20/D15.

Pin 27 is connected to GND.

Pin 28 is connected to +3V3.

Pin 29 is connected to GND.

A 100nF capacitor C40 is connected between pins 27 and 28.

The diagram shows the APS6404L-3SQRx-SN component (U18) with the following connections:

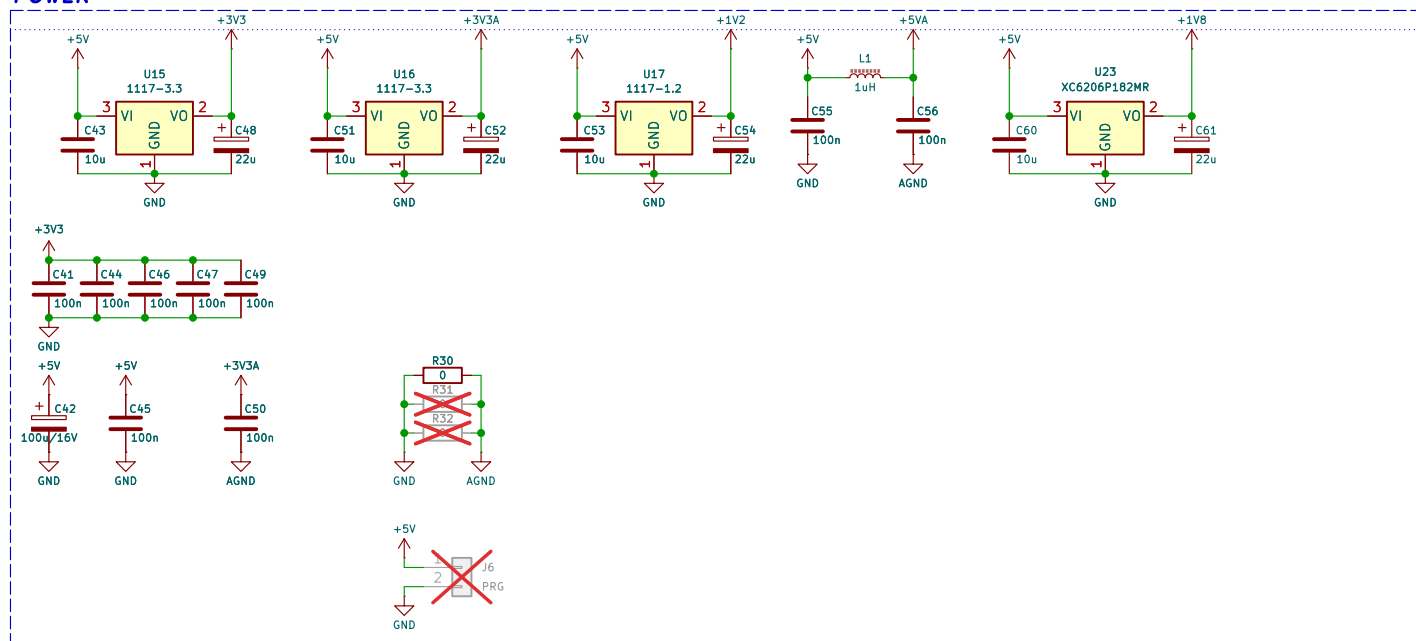
- Pin 1 (VDD):** Connected to +3V3 via resistor R49 (10k).
- Pin 2 (VSS):** Connected to GND.
- Pin 3 (SIO0):** Connected to +3V3.
- Pin 4 (SIO1):** Connected to GND.
- Pin 5 (SIO2):** Connected to +3V3.
- Pin 6 (SIO3):** Connected to GND.
- Pin 7 (SIO4):** Connected to +3V3.

A 100nF capacitor (C57) is connected between VDD and GND. The component is labeled U18 and APS6404L-3SQRx-SN.

[illegible]

Id: 6/8

POWER



Andy Karpov

Sheet: /Power/

File: power.kicad_sch

Title: Karabas-MegaBuzz

Size: A4

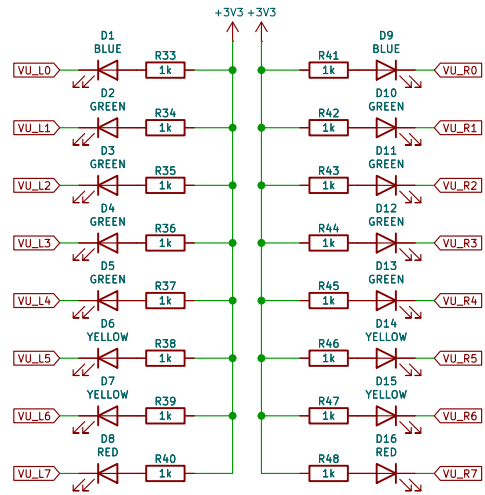
Date: 2026-09-19

Rev: A3

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Id: 7/8

VU METER



Andy Karpov

Sheet: /VU/

File: vu.kicad_sch

Title: Karabas-MegaBuzz

Size: A4

Date: 2026-09-19

Rev: A3

KiCad E.D.A. 10.0.6

Id: 8/8